TOWER OF THE MAXIMAL ABELIAN EXTENSIONS OF LOCAL FIELDS AND ITS APPLICATION

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1. Introduction

The local class field theory gives an explicit description of Galois groups of the abelian extensions of the local fields in terms of the ground fields. Langland suggested the extensive generalization of the local class field theory to the non-abelian extensions from the representation point of view, which is so called the local Langland's correspondence and was proved by M.Harris and R.Taylor in [5]. Nevertheless, since the Galois groups of local fields are always solvable, one can also study the Galois groups of local fields by filtering into abelian pieces. Indeed, Koch and de Shalit suggested this approach and studied the metabelian extensions by using the Coleman's power series in [6]. Fesenko further defined the non-commutative reciprocity cocycle in [3] (see also [4, Chapter IV, §8]). This paper can be regarded as the continuation of their work. However, we study the algebraic structure (with the discrete topology) of the principal units of infinite extensions by using Galois cohomology and recover the Galois groups by using the Kummer theory. This method allows us to study the higher level cases. Another motivation of this paper is to study the coverings of the maximal abelian extensions of local fields, which are the local analogue of [1]. This is a question raised by Linsheng Yin. As application, we prove the finiteness theorem for the coverings of the maximal abelian extensions of local fields. Moreover the explicit computation is given by using Fontaine's ϕ-Γ-module in [2].

It should be pointed out that the local fields we study in this paper is of characteristic 0 and almost all results in this paper are not true for the local fields of characteristic p.

Notation is standard if not explained. For any prime p, let $\mathbb{Q}_p$ be the p-adic numbers and K be a finite extension of $\mathbb{Q}_p$. Let k be the residue field of K and π_K be a uniformizer of K. Fix an algebraic closure $\bar{K}$ of K and use ord to denote the unique valuation over $\bar{K}$ with normalization $ord(\pi_K) = 1$ and $q = \sharp k$. Let K_{ur} be the maximal unramified extension of K and K_{tm} be the maximal tamely ramified extension of K. For any positive integer r, we use μ_r to denote the set of all r-th

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roots of unity. Write

$$\mu_{p^\infty} = \lim_{\rightarrow} \mu_{p^k} \quad \text{and} \quad \mathbb{Z}_p(1) = \lim_{\leftarrow} \mu_{p^k}$$

with respect to the natural inclusion and the p-th power map.

For a field L with $K \subseteq L \subseteq \bar{K}$, we write

$$\mathfrak{o}_L = \{x \in L : \quad ord(x) \geq 0\}$$

for the integral ring of L and

$$P_L = \{x \in L : \quad ord(x) > 0\}$$

for the maximal ideal of $\mathfrak{o}_L$ and $1 + P_L$ for the principal units of L. Since

$$(1 + P_L) = \lim_{\rightarrow_F} (1 + P_F)$$

for all finite extensions F/K inside L and each $(1 + P_F)$ is a $\mathbb{Z}_p$-module, one concludes that $(1 + P_L)$ is a $\mathbb{Z}_p$-module as well.

Definition 1.1. *Let $K^{(0)} = K$. We define that $K^{(n)}$ is the maximal abelian extension of $K^{(n-1)}$ for $n \geq 1$ inductively.*

The paper is organized as follows. We first study the tame extensions of K which is contained in $K^{(2)}$ in §2 and metabelian case in §3. Then $K^{(n)}$ for $n \geq 3$ is studied in §4. In §5, we prove the finiteness theorem for the coverings of $K^{(1)}$. Finally the explicit computation is given for $K = \mathbb{Q}_p$ in §6.

2. Tame case

We first deal with the tamely ramified extensions which are the most simplest case.

Lemma 2.1. *Suppose F is a field satisfying $K_{ur} \subseteq F \subseteq \bar{K}$. For any given positive integer m with $(m, p) = 1$, there is at most one finite extension E of F with $[E : F] = m$. Moreover E/F is a Kummer extension. Furthermore E/K is Galois if F/K is Galois.*

Proof. Suppose there are two finite extensions $F(\theta)/F$ and $F(\eta)/F$ of degree m with $(m, p) = 1$. Let

$$f(x) = x^m + a_1 x^{m-1} + \cdots + a_{m-1}x + a_m \text{ and } g(x) = x^m + b_1 x^{m-1} + \cdots + b_{m-1}x + b_m$$

where $a_1, \cdots, a_m$ and $b_1, \cdots, b_m \in F$ be the minimal polynomials of θ and η respectively. Let

$$L = K(a_1, \cdots, a_m, b_1, \cdots, b_m, \zeta_m)$$

where ζ_m is a primitive m-th root of unity. Then both $L(\theta)/L$ and $L(\eta)/L$ are tamely ramified of degree m. Since $(1 + P_L)$ is a $\mathbb{Z}_p$-module and m is invertible in $\mathbb{Z}_p$, $L(\theta)/L$ and $L(\eta)/L$ can be obtained by adjoining to L the roots of the equations $x^m = \pi_L$ and $x^m = \xi\pi_L$ respectively by [4] of Chap.II, (3.5) Proposition, where ξ is

a root of unity and π_L is a uniformizer of L. Since $F \supseteq K_{ur}$ which contains all roots of unity of order prime to p, there is $\zeta \in F$ such that $\xi = \zeta^m$. This implies that one get $F(\theta) = F(\eta)$ which are obtained by adjoining to F the roots of $x^m = \pi_L$ and $x^m = \xi\pi_L$ respectively. It is clear that E/F is a Kummer extension.

If F/K is Galois, then E/K is Galois by uniqueness of E with the fixed degree. □

Then K_{tm}/K_{ur} is a Kummer extension by Lemma 2.1 and $\bar{K}/K_{tm}$ is a pro-p-extension. Therefore $K_{tm} \subseteq K^{(2)}$ and $K^{(n)}/K^{(n-1)}$ is a Kummer pro-p-extension for $n \geq 3$.

Proposition 2.2. *For any positive integer m with $(m,p) = 1$, there is a unique field $K_m^{(1)}$ such that $[K_m^{(1)} : K^{(1)}] = m$. Moreover $K_m^{(1)} = K^{(1)}(\sqrt[m]{(\pi_K)^{\frac{1}{q-1}}})$*

Proof. The uniqueness follows from Lemma 2.1 by taking $F = K^{(1)}$. One only needs to show the existence of such field. Let $\delta = (\pi_K)^{\frac{1}{q-1}}$. Then δ is a uniformizer of $K_{tm} \cap K^{(1)}$ and $K^{(1)}/(K_{tm} \cap K^{(1)})$ is a pro-p-extension by the local class field theory and Chap.II, (3.5) Proposition of [4]. This implies that the polynomial $x^m - \delta$ is irreducible over $K_{tm} \cap K^{(1)}$ and remains irreducible over $K^{(1)}$. The proof is complete. □

In particular, there is the unique extension of degree $p-1$ over the maximal abelian extension of $\mathbb{Q}_p$ and is given by $\mathbb{Q}_p^{(1)}(\sqrt[p-1]{\zeta_p - 1})$ where ζ_p is a primitive p-th root of unity. It is clear that this field is Galois over $\mathbb{Q}_p$.

By modifying the arguments in Lemma 2.1, one has the following the finiteness result.

Corollary 2.3. *Let F be a field inside $\bar{K}$ and m be a positive integer with $(m,p) = 1$. Then the number of extensions of F with degree m is bounded by $\sigma_2(m)$ where $\sigma_2(m)$ is the sum of squares of all positive divisors of m.*

Proof. When F/K is a finite extension, any finite extension of F with degree m can be obtained by first the unique unramified extension $F_{m/d}$ of degree m/d and then some tamely ramified extension of degree d where d is a positive divisor of m. Since all tamely ramified extensions of $F_{m/d}$ of degree d can be given by adjoining a root of equation $x^d - \epsilon\pi_F$ to $F_{m/d}$ where π_F is the uniformizer of F and ϵ is a unit in $F_{m/d}$ by Chap.II (3.5) Proposition of [4], the number of tamely ramified extensions of $F_{m/d}$ with degree d is bounded by $d \cdot \#(\mathfrak{o}_{F_{m/d}}^\times/(\mathfrak{o}_{F_{m/d}}^\times)^d)$. Since $(d,p) = 1$, one has

$$\#(\mathfrak{o}_{F_{m/d}}^\times/(\mathfrak{o}_{F_{m/d}}^\times)^d) = \#[(\mathfrak{o}_{F_{m/d}}/P_{F_{m/d}})^\times/((\mathfrak{o}_{F_{m/d}}/P_{F_{m/d}})^\times)^d] = d.$$

Therefore the number of all extensions of F with degree m is bounded by $\sigma_2(m) = \sum_{d|m} d^2$ which is independent of F.

In general, let $F(\theta_i)$ be the extensions of F of degree m with $1 \leq i \leq \sigma_2(m)+1$. We will show at least two of them coincide. Let L be a finite extension K inside F obtained by adjoining all coefficients of the minimal polynomials of θ_i over F to K for $1 \leq i \leq \sigma_2(m)+1$. Then $L(\theta_i)$ for $1 \leq i \leq \sigma_2(m)+1$ are the extensions of L of degree m. By the above argument, there are $1 \leq i_0, j_0 \leq \sigma_2(m)+1$ such that $L(\theta_{i_0}) = L(\theta_{j_0})$. Therefore $F(\theta_{i_0}) = F(\theta_{j_0})$. □

3. Metabelian case

Since

$$ord(1-\zeta_{p^k}) = \frac{ord(p)}{(p-1)p^{k-1}}$$

for any primitive p^k-th root of unity ζ_{p^k}, one has μ_{p^∞} is a discrete torsion $\mathbb{Z}_p$-module in the p-adic topology. For any field extension E/K, one writes the maximal quotient of torsion free $\mathbb{Z}_p$-module of $(1+P_E)$ as

$$\mathfrak{I}_E = (1+P_E)/(\mu_{p^\infty} \cap E).$$

Theorem 3.1. *If E/K is abelian, then $\mathfrak{I}_E$ is a free $\mathbb{Z}_p$-module. Moreover $\mathfrak{I}_E$ is of countable infinite rank if and only if $[E:K] = \infty$.*

Proof. By [4] of Chap.I §6, one only needs to show that $\mathfrak{I}_E$ is of countable infinite rank if $[E:K] = \infty$. Write $\mu(E) = \mu_{p^\infty} \cap E$.

Let L be a finite abelian extension of K inside E and $G_L = \mathrm{Gal}(E/L)$. Applying the Galois cohomology to the short exact sequence

$$1 \to \mu(E) \to (1+P_E) \to \mathfrak{I}_E \to 1,$$

one has the long exact sequence

$$1 \to \mu_{p^l} \to (1+P_L) \to (\mathfrak{I}_E)^{G_L} \to H^1(G_L, \mu(E)) \to \cdots$$

where μ_{p^l} is the set of all $p-th$ power roots of unity contained in L.

We claim that $H^1(G_L, \mu(E))$ is finite. By the inflation-restriction sequence, one only needs to prove the claim is true for a finite extension of L. Therefore one can assume that $Gal[L(\mu(E))/L]$ is pro-cyclic extension. Let $H_L = Gal[E/L(\mu(E))]$. By the inflation-restriction sequence, one has the exact sequence

$$1 \to H^1(Gal[L(\mu(E))/L], \mu(E)) \to H^1(G_L, \mu(E)) \to H^1(H_L, \mu(E))^{G_L/H_L}.$$

Since H_L acts on $\mu(E)$ trivially, one has

$$\begin{aligned} H^1(H_L, \mu(E))^{G_L/H_L} &= Hom_{cont}(H_L, \mu(E))^{Gal[L(\mu(E))/L]} \\ &= Hom_{cont}(H_L, \mu_{p^l}) \subseteq Hom_{\mathbb{Z}_p}(H_L(p), \mu_{p^l}) \end{aligned}$$

where $H_L(p)$ is the pro-p-part of H_L. By the local class field theory, $H_L(p)$ is the sub-quotient $\mathbb{Z}_p$-module of $(1+P_K)\times\mathbb{Z}_p$ which is a finitely generated by [4] of Chap.I

(6.5) Corollary (3). This implies that $H_L(p)$ is a finitely generated $\mathbb{Z}_p$-module as well. Therefore $Hom_{\mathbb{Z}_p}(H_L(p), \mu_{p^l})$ is finite. Moreover

$$H^1(Gal[L(\mu(E))/L], \mu(E)) = \lim_{\to_n} H^1(Gal[L(\mu_{p^n} \cap E)/L], (\mu_{p^n} \cap E))$$

and $Gal[L(\mu_{p^n} \cap E)/L]$ is cyclic, one has

$$\#H^1(Gal[L(\mu_{p^n} \cap E)/L], \mu_{p^n}) = \#\hat{H}^0(Gal[L(\mu_{p^n} \cap E)/L], \mu_{p^n}) \leq p^l$$

by [8] of Chap.VIII, §4, Prop.8. Therefore

$$\#H^1(Gal[L(\mu(E))/L], \mu(E)) \leq p^l$$

and the claim follows.

Since $(\mathfrak{I}_E)^{G_L}$ is torsion-free, one concludes that $(\mathfrak{I}_E)^{G_L}$ is a free $\mathbb{Z}_p$-module of rank $[L : \mathbb{Q}_p]$ by the above claim and by [4] of Chap.I (6.5) Corollary (3). Moreover, suppose there are $m \in \mathbb{Z}_p$ and $a \in \mathfrak{I}_E$ such that $a^m \in (\mathfrak{I}_E)^{G_L}$. Then $\sigma a/a$ is a torsion element in $\mathfrak{I}_E$ for all $\sigma \in G_L$. Since $\mathfrak{I}_E$ is torsion free, one has $a \in (\mathfrak{I}_E)^{G_L}$. This implies that $\mathfrak{I}_E/(\mathfrak{I}_E)^{G_L}$ is torsion free.

Let L' be a finite abelian extension of K with $E \supseteq L' \supseteq L$ and $G_{L'} = Gal(E/L')$. Then

$$(\mathfrak{I}_E)^{G_L} \subseteq (\mathfrak{I}_E)^{G_{L'}}$$

and the quotient

$$(\mathfrak{I}_E)^{G_{L'}}/(\mathfrak{I}_E)^{G_L}$$

is torsion free and therefore is a free $\mathbb{Z}_p$-module. This implies that $(\mathfrak{I}_E)^{G_L}$ is a direct summand of $(\mathfrak{I}_E)^{G_{L'}}$ as $\mathbb{Z}_p$-module. Since

$$\mathfrak{I}_E = \lim_{\to_L} (\mathfrak{I}_E)^{G_L},$$

one concludes that $\mathfrak{I}_E$ is a free $\mathbb{Z}_p$-module of countable infinite rank. □

Remark 3.2. *M may not be a free $\mathbb{Z}_p$-module in general if M is a torsion-free and $M \otimes \mathbb{Q}_p$ is a finite dimensional $\mathbb{Q}_p$-vector space and any non-zero element in M is not p-divisible. For example, let*

$$M = (\oplus_{i=0}^{\infty} \mathbb{Z}_p x_i)/S$$

where S is $\mathbb{Z}_p$-module generated by $\{x_0 + x_i - px_{i+1} | i \geqslant 1\}$. One can verify that M satisfies the above properties but is not free.

Lemma 3.3. *Suppose $E \supseteq K(\mu_{p^\infty})$. Then the projection map induces the isomorphism*

$$E^\times/(E^\times)^{p^n} \cong (1 + P_E)/(1 + P_E)^{p^n}$$

for any positive integer n.

Proof. For any $x \in E^{\times}$, there are $s, m \in \mathbb{Z}$ with $(m, s) = 1$ such that $ord(x) = \frac{s}{m}$. We can assume that $(m, p) = 1$, otherwise one can consider $ord(x^{-1})$ instead. Then there are $u, v \in \mathbb{Z}$ such that $um + vp^n = 1$. Therefore $ord(\frac{x}{(x^v)^{p^n}}) \in \mathbb{Z}$. So we can assume $ord(x) \in \mathbb{Z}$. By the ramification argument, there is a positive integer N such that $K(\mu_{p^\infty})/K(\mu_{p^N})$ is a totally ramified $\mathbb{Z}_p$-extension. If y is a uniformizer of $K(\mu_{p^{N+n}})$, then

$$1 = ord(\pi_K) = ord((y^e)^{p^n})$$

where e is the ramification index of $K(\mu_{p^N})/K$. Therefore we can further assume that $ord(x) = 0$. The result follows from the perfectness of the residue field of E. □

By Theorem 3.1, one has that $(1 + P_{K^{(1)}})$ is the direct sum of the torsion part with the free part as $\mathbb{Z}_p$-module. Let $B_{K^{(1)}}$ be the set of $\mathbb{Z}_p$-basis of the free part of $(1 + P_{K^{(1)}})$.

Corollary 3.4. *Let*

$$K_t^{(2)} = K^{(1)} \cdot K_{tm} = \cup_{(s,p)=1} K^{(1)}(\sqrt[s]{(\pi_K)^{\frac{1}{q-1}}})$$

and

$$K_w^{(2)} = \cup_{n=1}^{\infty} K^{(1)}(\sqrt[p^n]{B_{K^{(1)}}}).$$

Then

$$K^{(2)} = K_t^{(2)} \cdot K_w^{(2)} \quad with \quad K_t^{(2)} \cap K_w^{(2)} = K^{(1)}.$$

Moreover

$$Gal(K_t^{(2)}/K^{(1)}) \cong Gal[K_{tm}/(K^{(1)} \cap K_{tm})] \cong \prod_{l \neq p} \mathbb{Z}_l$$

where l's are primes and

$$Gal(K_w^{(2)}/K^{(1)}) \cong Hom_{\mathbb{Z}_p}(\mathfrak{I}_{K^{(1)}}, \mathbb{Z}_p(1)) \cong \prod_{B^{(1)}} \mathbb{Z}_p.$$

Proof. Since any finite abelian group can be decomposed as the direct sum of the p-part and the non-p-part, one obtains that any non-p-part finite abelian extension of $K^{(1)}$ is given by

$$K^{(1)}(\sqrt[s]{(\pi_K)^{\frac{1}{q-1}}}) \subseteq K^{(1)} \cdot K_{tm}$$

with some positive integer s by Prop.2.2. Therefore $K_t^{(2)}$ is the union of all finite abelian non-p-part extension of $K^{(1)}$ with

$$K_t^{(2)} = K^{(1)} \cdot K_{tm} = \cup_{(s,p)=1} K^{(1)}(\sqrt[s]{(\pi_K)^{\frac{1}{q-1}}})$$

and

$$Gal(K_t^{(2)}/K^{(1)}) \cong \prod_{l \neq p} \mathbb{Z}_l$$

by [4] of Chap.II, (3.5) Proposition and Prop.2.2 and the Kummer theory.

Since $K^{(1)}$ contains all roots of unity, the p-part of any finite abelian extension of $K^{(1)}$ is a Kummer extension as well. By Lemma 3.3 and Thm.3.1, any finite abelian p-extension of $K^{(1)}$ is contained in $K^{(1)}(\sqrt[p^n]{B_{K^{(1)}}})$ for some positive integer n and

$$\begin{aligned} Gal(K_w^{(2)}/K^{(1)}) &= \lim_{\leftarrow} Gal[K^{(1)}(\sqrt[p^n]{B_{K^{(1)}}})/K^{(1)}] \\ &\cong \lim_{\leftarrow} Hom_{\mathbb{Z}_p}(\mathfrak{I}_{K^{(1)}}/(\mathfrak{I}_{K^{(1)}})^{p^n}, \mu_{p^n}) \\ &\cong \lim_{\leftarrow} Hom_{\mathbb{Z}_p}(\mathfrak{I}_{K^{(1)}}, \mu_{p^n}) \cong Hom_{\mathbb{Z}_p}(\mathfrak{I}_{K^{(1)}}, \mathbb{Z}_p(1)) \cong \prod_{B_{K^{(1)}}} \mathbb{Z}_p \end{aligned}$$

by the Kummer theory. It is clear that

$$K^{(2)} = K_t^{(2)} \cdot K_w^{(2)} \quad \text{and} \quad K_t^{(2)} \cap K_w^{(2)} = K^{(1)}$$

by the above construction. The proof is complete. □

Corollary 3.5. *Let $E \supseteq K(\mu_{p^\infty})$ and E_p be the maximal abelian pro-p-extension of E. If E/K is abelian, then*

$$Gal(E_p/E) \cong Hom_{\mathbb{Z}_p}(\mathfrak{I}_E, \mathbb{Z}_p(1)).$$

Proof. It follows from the same arguments of Corollary 3.4 and Lemma 3.3. □

4. Higher level cases

Let E/F be a field extension inside $\bar{K}/K$.

Definition 4.1.

$$\mathfrak{r}_{E/F} = \{x \in (1+P_E) : \ x^{p^m} \in (1+P_F) \ \ \textit{for some positive integer } m\}.$$

Then

$$(1+P_F) \subset \mathfrak{r}_{E/F} \subset (1+P_E).$$

By Krasner's lemma, $(1+P_F)$ is the closed submodule of $(1+P_E)$ in the p-adic topology.

Proposition 4.2. *If E/F be a field extension inside $\bar{K}/K$, then $\mathfrak{r}_{E/F}$ is the closed $\mathbb{Z}_p$-submodule of $(1+P_E)$ in the p-adic topology.*

Proof. Let $x_0 \in E$ be the limit of the Cauchy sequence $\{x_n\}_{n=1}^\infty$ contained in $\mathfrak{r}_{E/F}$ and D/F be the Galois closure of E/F inside $\bar{K}$. Then $\{\sigma x_n\}_{n=1}^\infty$ is also a Cauchy sequence with the limit $\sigma x_0 \in D$ for any $\sigma \in Gal(D/F)$. Therefore

$$\lim_{n\to\infty} \frac{\sigma x_n}{x_n} = \frac{\sigma x_0}{x_0}.$$

Since $\{x_n\}_{n=1}^\infty \subset \mathfrak{r}_{E/F}$, one has

$$\{(\frac{\sigma x_n}{x_n})\}_{n=1}^\infty \subseteq \mu(E).$$

It is clear that $\mu(E)$ is discrete, hence is closed. Therefore $(\sigma x_0/x_0) \in \mu(E)$. The result follows from the finiteness of conjugate elements of x_0 over F. □

Define

$$I_{E/F} = (1 + P_E)/\mathfrak{r}_{E/F}.$$

The main result of this section is the following theorem.

Theorem 4.3. *Suppose both E/K and F/K are Galois and $F \supseteq K(\mu_{p^\infty})$. If E/F is a non-trivial abelian extension, then $I_{E/F}$ is a free $\mathbb{Z}_p$-module of countable infinite rank.*

Proof. Let $L'/L/K$ be two finite extensions inside E. One can verify directly that the natural inclusion induces the injective homomorphism

$$1 \to \mathfrak{r}_{E/L}/\mathfrak{r}_{E/F\cap L} \to \mathfrak{r}_{E/L'}/\mathfrak{r}_{E/F\cap L'}$$

and the quotient

$$\frac{\mathfrak{r}_{E/L'}/\mathfrak{r}_{E/F\cap L'}}{\mathfrak{r}_{E/L}/\mathfrak{r}_{E/F\cap L}} \cong \frac{\mathfrak{r}_{E/L'}}{\mathfrak{r}_{E/L}}$$

is a torsion free $\mathbb{Z}_p$-module by Definition 4.1. Since E is the union of all finite extensions of K inside E, one has

$$I_{E/F} = \varinjlim_{L} \mathfrak{r}_{E/L}/\mathfrak{r}_{E/F\cap L}$$

where L runs over all finite extensions of K inside E. It is sufficient to show that $\mathfrak{r}_{E/L}/\mathfrak{r}_{E/F\cap L}$ is a free $\mathbb{Z}_p$-module of finite rank for any finite extension L/K.

First we descend from E to FL. Let $H_L = Gal(E/FL)$. By the short exact sequence commutative diagram of natural inclusion

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mu_{p^\infty} & \longrightarrow & \mathfrak{r}_{E/F\cap L} & \longrightarrow & \mathfrak{r}_{E/F\cap L}/\mu_{p^\infty} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \mu_{p^\infty} & \longrightarrow & \mathfrak{r}_{E/L} & \longrightarrow & \mathfrak{r}_{E/L}/\mu_{p^\infty} & \longrightarrow & 1, \end{array}$$

one obtains the following long exact sequence commutative diagram by using Galois cohomology

$$\begin{array}{ccccccccccc} 1 & \longrightarrow & \mathfrak{r}_{FL/F\cap L}/\mu_{p^\infty} & \longrightarrow & (\mathfrak{r}_{E/F\cap L}/\mu_{p^\infty})^{H_L} & \longrightarrow & H^1(H_L, \mu_{p^\infty}) & \xrightarrow{\xi} & H^1(H_L, \mathfrak{r}_{E/F\cap L})\cdots \\ & & \downarrow & & \downarrow & & \downarrow{\scriptstyle id} & & \downarrow \\ 1 & \longrightarrow & \mathfrak{r}_{FL/L}/\mu_{p^\infty} & \longrightarrow & (\mathfrak{r}_{E/L}/\mu_{p^\infty})^{H_L} & \longrightarrow & H^1(H_L, \mu_{p^\infty}) & \xrightarrow{\eta} & H^1(H_L, \mathfrak{r}_{E/L})\cdots \end{array}$$

Since

$$(\mathfrak{r}_{E/F\cap L}/\mu_{p^\infty})^{H_L} = \mathfrak{r}_{E/F\cap L}/\mu_{p^\infty} \quad \text{and} \quad (\mathfrak{r}_{E/L}/\mu_{p^\infty})^{H_L} = \mathfrak{r}_{E/L}/\mu_{p^\infty},$$

one has the following short exact sequence from the above long exact sequence

$$1 \to \mathfrak{r}_{FL/L}/\mathfrak{r}_{FL/F\cap L} \to \mathfrak{r}_{E/L}/\mathfrak{r}_{E/F\cap L} \to ker(\eta)/ker(\xi) \to 1.$$

Let

$$N = res^{Gal(E/F)}_{H_L} \circ cor^{H_L}_{Gal(E/F)} : \ H^1(H_L, \mu_{p^\infty}) \to H^1(H_L, \mu_{p^\infty}).$$

Then N is the norm map by [7] of (1.5.7) Corollary. Since the maps *cor* and *res* commute with the connection map by [7] of (1.5.2) Proposition, one has

$$N(ker(\eta)) \subseteq ker(\xi).$$

Moreover H_L acts on μ_{p^∞} trivially and $Gal(E/F)$ is abelian, N is the same as the multiplication by $[FL : F]$. Therefore $ker(\eta)/ker(\xi)$ is killed by $[FL : F]$. It can be verified directly by Definition 4.1 that both $\mathfrak{r}_{FL/L}/\mathfrak{r}_{FL/F\cap L}$ and $\mathfrak{r}_{E/L}/\mathfrak{r}_{E/F\cap L}$ are torsion free $\mathbb{Z}_p$-module. One concludes that $\mathfrak{r}_{FL/L}/\mathfrak{r}_{FL/F\cap L}$ is a free $\mathbb{Z}_p$-module of finite rank if and only if $\mathfrak{r}_{E/L}/\mathfrak{r}_{E/F\cap L}$ is a free $\mathbb{Z}_p$-module of finite rank. In this case, these two $\mathbb{Z}_p$-modules have the same rank.

Let $G_L = Gal(FL/L)$ and consider the natural inclusion of the short exact sequence commutative diagram

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \mu_{p^\infty} & \longrightarrow & \mathfrak{r}_{FL/F} & \longrightarrow & \mathfrak{r}_{FL/F}/\mu_{p^\infty} & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \mu_{p^\infty} & \longrightarrow & (1+P_{FL}) & \longrightarrow & (1+P_{FL})/\mu_{p^\infty} & \longrightarrow & 1.
\end{array}$$

One obtains the following long exact sequence commutative diagram by using Galois cohomology

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \frac{\mathfrak{r}_{L/L\cap F}}{\mu_{p^l}} & \longrightarrow & (\frac{\mathfrak{r}_{FL/F}}{\mu_{p^\infty}})^{G_L} & \longrightarrow & H^1(G_L,\mu_{p^\infty}) & \xrightarrow{\phi} & H^1(G_L,\mathfrak{r}_{FL/F})\cdots \\
 & & \downarrow & & \downarrow & & \downarrow{\scriptstyle id} & & \downarrow \\
1 & \longrightarrow & \frac{1+P_L}{\mu_{p^l}} & \longrightarrow & (\frac{1+P_{FL}}{\mu_{p^\infty}})^{G_L} & \longrightarrow & H^1(G_L,\mu_{p^\infty}) & \xrightarrow{\psi} & H^1(G_L,(1+P_{FL}))\cdots
\end{array}$$

where $\mu_{p^l} = \mu_{p^\infty} \cap L$. It can be verified directly by Definition 4.1 that

$$\mathfrak{r}_{FL/F\cap L}/\mu_{p^\infty} = (\mathfrak{r}_{FL/F}/\mu_{p^\infty})^{G_L} \quad \text{and} \quad \mathfrak{r}_{FL/L}/\mu_{p^\infty} = ((1+P_{FL})/\mu_{p^\infty})^{G_L}.$$

Then the following short exact sequence follows from the above long exact sequence commutative diagram

$$1 \to (1+P_L)/\mathfrak{r}_{L/F\cap L} \to \mathfrak{r}_{FL/L}/\mathfrak{r}_{FL/F\cap L} \to ker(\psi)/ker(\phi) \to 1.$$

By using the norm map $N_{FL/F}$ and the natural inclusion, one has the following short exact sequence commutative diagram

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \mu_{p^\infty} & \longrightarrow & (1+P_{FL}) & \longrightarrow & (1+P_{FL})/\mu_{p^\infty} & \longrightarrow & 1 \\
 & & {\scriptstyle [FL:F]}\downarrow & & {\scriptstyle N_{FL/F}}\downarrow & & \downarrow & & \\
1 & \longrightarrow & \mu_{p^\infty} & \longrightarrow & (1+P_F) & \longrightarrow & (1+P_F)/\mu_{p^\infty} & \longrightarrow & 1 \\
 & & {\scriptstyle id}\downarrow & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \mu_{p^\infty} & \longrightarrow & \mathfrak{r}_{FL/F} & \longrightarrow & \mathfrak{r}_{FL/F}/\mu_{p^\infty} & \longrightarrow & 1.
\end{array}$$

By the Galois cohomology, one obtains the following commutative diagram

$$\begin{array}{ccc} H^1(G_L,\mu_{p^\infty}) & \xrightarrow{\psi} & H^1(G_L,(1+P_{FL})) \\ {\scriptstyle [FL:F]}\downarrow & & {\scriptstyle N_{FL/F}}\downarrow \\ H^1(G_L,\mu_{p^\infty}) & \longrightarrow & H^1(G_L,(1+P_F)) \\ {\scriptstyle id}\downarrow & & \downarrow \\ H^1(G_L,\mu_{p^\infty}) & \xrightarrow{\phi} & H^1(G_L,\mathfrak{r}_{FL/F}). \end{array}$$

Therefore $[FL:F]ker(\psi) \subseteq ker(\phi)$. By Definition 4.1, one can verify directly that both $\mathfrak{r}_{FL/L}/\mathfrak{r}_{FL/F\cap L}$ and $(1+P_L)/\mathfrak{r}_{L/F\cap L}$ are the torsion free $\mathbb{Z}_p$-module. Then $(1+P_L)/\mathfrak{r}_{L/F\cap L}$ is a free $\mathbb{Z}_p$-module of finite rank by [4] of Chap.I (6.5) Corollary (3). Therefore one concludes that $\mathfrak{r}_{FL/L}/\mathfrak{r}_{FL/F\cap L}$ is a free $\mathbb{Z}_p$-module of finite rank.

Moreover $\mathfrak{r}_{L/F\cap L}/(1+P_{F\cap L})$ is a torsion $\mathbb{Z}_p$-module, one has

$$rank(\mathfrak{r}_{L/F\cap L}) = rank(1+P_{F\cap L}).$$

Therefore, by the above arguments and [4] of Chap.I (6.5) Corollary (3),

$$\begin{aligned} &rank(\mathfrak{r}_{E/L}/\mathfrak{r}_{E/F\cap L}) = rank(\mathfrak{r}_{FL/L}/\mathfrak{r}_{FL/F\cap L}) \\ =&rank(1+P_L) - rank(1+P_{F\cap L}) \\ =&([L:F\cap L]-1)[F\cap L:\mathbb{Q}_p] \end{aligned}$$

which goes to infinity when L is big enough inside E since $[F:\mathbb{Q}_p]$ is infinite. The proof is complete. □

Let

$$I_K^{(n)} = I_{K^{(n)}/K^{(n-1)}} = (1+P_{K^{(n)}})/\mathfrak{r}_{K^{(n)}/K^{(n-1)}}$$

for $n \geq 2$. One has the following result about the algebraic structure of the principal units $(1+P_{K^{(n)}})$ of $K^{(n)}$.

Corollary 4.4. *For $n \geq 2$,*

$$(1+P_{K^{(n)}}) \cong \mathfrak{r}_{K^{(n)}/K^{(n-1)}} \times I_K^{(n)}$$

where $I_K^{(n)}$ is a free $\mathbb{Z}_p$-module of countable infinite rank and $\mathfrak{r}_{K^{(n)}/K^{(n-1)}}$ is p-divisible such that $\mathfrak{r}_{K^{(n)}/K^{(n-1)}}/\mu_{p^\infty}$ is a countable infinite dimensional $\mathbb{Q}_p$-vector space.

Proof. It is clear that $K^{(n)}/K$ is Galois for all $n \geq 2$ by the maximality. Since $K^{(2)}/K^{(1)}$ is a non-trivial abelian extension by Corollary 3.4, one has that $I_K^{(2)}$ is a free $\mathbb{Z}_p$-module of countable infinite rank by Thm. 4.3.

Since $K^{(n)}$ is the maximal abelian extension of $K^{(n-1)}$ and $\mu_{p^\infty} \subseteq K^{(n-1)}$, one has $K^{(n)}/K^{(n-1)}$ is a Kummer pro-p-extension for $n \geq 3$ by Lemma 2.1. Moreover

$\mathfrak{r}_{K^{(n)}/K^{(n-1)}}$ is p-divisible. This makes $\mathfrak{r}_{K^{(n)}/K^{(n-1)}}/\mu_{p^\infty}$ as a countable infinite dimensional $\mathbb{Q}_p$-vector space. Since

$$\begin{aligned}\frac{K^{(n-1)^\times}}{(K^{(n-1)^\times})^{p^k}} &\cong \frac{(1+P_{K^{(n-1)}})}{(1+P_{K^{(n-1)}})^{p^k}}\\ \cong \frac{\mathfrak{r}_{K^{(n-1)}/K^{(n-2)}}}{(\mathfrak{r}_{K^{(n-1)}/K^{(n-2)}})^{p^k}} &\times \frac{I_K^{(n-1)}}{(I_K^{(n-1)})^{p^k}} \cong \frac{I_K^{(n-1)}}{(I_K^{(n-1)})^{p^k}}\end{aligned}$$

by Lemma 3.3 and the induction hypothesis, one concludes that $K^{(n)}$ is a non-trivial abelian extension of $K^{(n-1)}$ and $I_K^{(n)}$ is a free $\mathbb{Z}_p$-module of countable infinite rank by Thm. 4.3. The proof is complete. □

Let $B_{K^{(n)}}$ be a $\mathbb{Z}_p$-basis of the free part of $(1+P_{K^{(n)}})$.

Corollary 4.5. *For $n \geq 2$,*

$$K^{(n+1)} = \cup_{k=1}^{\infty} K^{(n)}(\sqrt[p^k]{B_{K^{(n)}}})$$

with

$$Gal(K^{(n+1)}/K^{(n)}) \cong Hom_{\mathbb{Z}_p}(I_K^{(n)}, \mathbb{Z}_p(1)) \cong \prod_{B_{K^{(n)}}} \mathbb{Z}_p.$$

Proof. The first part follows from the same argument as that in Corollary 4.4. Moreover

$$\begin{aligned}Gal(K^{(n+1)}/K^{(n)}) &\cong \varprojlim_k Hom_{\mathbb{Z}_p}(I_K^{(n)}/(I_K^{(n)})^{p^k}, \mu_{p^k})\\ \cong \varprojlim_k Hom_{\mathbb{Z}_p}(I_K^{(n)}, \mu_{p^k}) &\cong Hom_{\mathbb{Z}_p}(I_K^{(n)}, \mathbb{Z}_p(1)) \cong \prod_{B_{K^{(n)}}} \mathbb{Z}_p\end{aligned}$$

by the Kummer theory. The proof is complete.

□

5. Finite coverings of abelian extensions of local fields

In this section, we will give an application of studying the local analogy of [1] by using the results in the previous sections. We keep the same notation as before. In particular, K is a finite extension of $\mathbb{Q}_p$ and $\bar{K}$ is a fixed algebraic closure of K.

Definition 5.1. *Let E/K be a field extension inside $\bar{K}/K$. A finite extension F/E of degree k is called a k-covering of E with respect to K if F/K is Galois.*

We will show the finiteness result of m-coverings for any positive integer m if E/K is abelian. First we need the following lemma.

Lemma 5.2. *Suppose $E \supseteq K(\mu_{p^\infty})$ and E/K is abelian. Let $G = Gal(E/K)$. Then $[\mathfrak{I}_E/(\mathfrak{I}_E)^p]^G$ is finite.*

Proof. By the Hochschild-Serre spectral sequence (see (1.6.6) Proposition of [7]), one has the following exact sequence

$$H^1(Gal(\bar{K}/K),\mu_p)\to H^1(Gal(\bar{K}/E),\mu_p)^G\to H^2(G,\mu_p).$$

Since

$$\begin{aligned}&H^1(Gal(\bar{K}/E),\mu_p)=Hom_{cont}(Gal(\bar{K}/E),\mu_p)\\ =&Hom_{cont}(Gal(E_p/E),\mu_p)\\ \cong&Hom_{cont}(Hom_{\mathbb{Z}_p}(\mathfrak{I}_E,\mathbb{Z}_p(1)),\mu_p)\cong\mathfrak{I}_E/(\mathfrak{I}_E)^p\end{aligned}$$

by Corollary 3.5 and the Kummer theory where E_p is the maximal abelian pro-p-extension of E and

$$H^1(Gal(\bar{K}/K),\mu_p)\cong K^\times/(K^\times)^p$$

which is finite, it is sufficient to show that $H^2(G,\mu_p)$ is finite.

First we can assume that $\mu_p\subset K$. Otherwise, we know that $[K(\mu_p):K]$ is a divisor of $[\mathbb{Q}_p(\mu_p):\mathbb{Q}_p]=p-1$ and the map

$$H^2(G,\mu_p)\xrightarrow{res}H^2(Gal[E/K(\mu_p)],\mu_p)\xrightarrow{cor}H^2(G,\mu_p)$$

is the multiplication of $[K(\mu_p):K]$ by [7] of (1.5.7) Corollary. Since $H^2(G,\mu_p)$ is p-torsion and one concludes that the finiteness of $H^2(Gal[E/K(\mu_p)],\mu_p)$ implies the finiteness of $H^2(G,\mu_p)$. Therefore one can make such assumption.

Let $C=Gal(K^{(1)}/E)$ and $C(p)$ be the pro-p-part of C. By the Hochschild-Serre spectral sequence (see (1.6.6) Proposition of [7]), one obtains the following exact sequence

$$H^1(C,\mu_p)^G\to H^2(G,\mu_p)\to H^2(Gal(K^{(1)}/K),\mu_p).$$

Since the pro-p-part of $Gal(K^{(1)}/K)$ is a finite generated $\mathbb{Z}_p$-module by the local class field theory, one obtains that the sub-$\mathbb{Z}_p$-module $C(p)$ is also finitely generated. Then

$$H^1(C,\mu_p)^G=Hom_{cont}(C,\mu_p)=Hom_{cont}(C(p),\mu_p)\subseteq Hom_{\mathbb{Z}_p}(C(p),\mu_p)$$

is finite. Therefore the finiteness of $H^2(Gal(K^{(1)}/K),\mu_p)$ implies that the finiteness of $H^2(G,\mu_p)$. One can further assume that $E=K^{(1)}$.

By the local class field theory, one has $G=\hat{\mathbb{Z}}\times\mathbb{Z}_p^r\times F$ where $r=[K:\mathbb{Q}_p]$ and F is a finite abelian group. Let $F(p)$ be the p-part of F. By the Hochschild-Serre spectral sequence, one obtains the following exact sequence

$$H^2(\hat{\mathbb{Z}},\mu_p)\to\ker[H^2(G,\mu_p)\to H^2(\mathbb{Z}_p^r\times F,\mu_p)^{\hat{\mathbb{Z}}}]\to H^1(\hat{\mathbb{Z}},H^1(\mathbb{Z}_p^r\times F,\mu_p)).$$

Since $H^2(\hat{\mathbb{Z}},\mu_p)=0$ by [7] of (1.6.13) Proposition and

$$\begin{aligned}&H^1(\hat{\mathbb{Z}},H^1(\mathbb{Z}_p^r\times F,\mu_p))=Hom_{cont}(\hat{\mathbb{Z}},Hom_{cont}(\mathbb{Z}_p^r\times F,\mu_p))\\ =&Hom_{cont}(\mathbb{Z}_p,Hom_{cont}(\mathbb{Z}_p^r\times F(p),\mu_p))\\ \subseteq&Hom_{\mathbb{Z}_p}(\mathbb{Z}_p,Hom_{\mathbb{Z}_p}(\mathbb{Z}_p^r\times F(p),\mu_p))\end{aligned}$$

is finite, one concludes that the finiteness of $H^2(\mathbb{Z}_p^r \times F, \mu_p)$ implies the finiteness of $H^2(G, \mu_p)$. By the Hochschild-Serre spectral sequence again, one obtains the following exact sequence

$$H^2(\mathbb{Z}_p, \mu_p) \to \ker[H^2(\mathbb{Z}_p^r \times F, \mu_p) \to H^2(\mathbb{Z}_p^{r-1} \times F, \mu_p)^{\mathbb{Z}_p}] \to H^1(\mathbb{Z}_p, H^1(\mathbb{Z}_p^{r-1} \times F, \mu_p)).$$

By the same arguments as above, one has that the finiteness of $H^2(\mathbb{Z}_p^{r-1} \times F, \mu_p)$ implies the finiteness of $H^2(\mathbb{Z}_p^r \times F, \mu_p)$. By induction on r, the result follows from the finiteness of $H^2(F, \mu_p)$. □

Now we can prove the finiteness result about the finite coverings over abelian extensions.

Theorem 5.3. *The number of k-coverings of E with respect to K is finite for any given positive integer k if E/K is abelian.*

Proof. Step 1. We first prove that this theorem is true when E contains μ_{p^∞}. For any positive integer r, we claim that there are only finitely many p^r-coverings F of E with respect to K such that $Gal(F/E)$'s are the elementary abelian p-groups. Indeed, such F's are obtained by several composition of $E(\sqrt[p]{x})$ for some representatives x in $\mathfrak{I}_E/(\mathfrak{I}_E)^p$ by the Kummer theory. Since F's are Galois over K, the numbers of elements in the orbit of x's in $\mathfrak{I}_E/(\mathfrak{I}_E)^p$ under the natural action of $Gal(E/K)$ are bounded by p^r by the Kummer theory. Then the indices of the stabilizers S_x of such x's in $Gal(E/K)$ are bounded by p^r as well. Since E/K is abelian, the number of open subgroups of indices bounded by p^r in $Gal(E/K)$ is bounded by the number of subgroups of $K^\times/(K^\times)^{p^r}$ by the local class field theory, which depends only on K and p^r. Therefore the number of S_x's is bounded by a number depending only on K and p^r. Since $[\mathfrak{I}_E/(\mathfrak{I}_E)^p]^{S_x}$ is finite by Lemma 5.2 and x is in $[\mathfrak{I}_E/(\mathfrak{I}_E)^p]^{S_x}$, the claim follows.

Let $k = tp^s$ with $(t, p) = 1$. By Corollary 2.3, there is a finite set $\mathcal{T}(E)$ of the extensions of E with the degree dividing t. By the above claim there is a finite set $\mathcal{F}(E)$ of all covering F's of E with respect to K such that $Gal(F/E)$ is an elementary abelian p-group and $\#Gal(F/E) \leq p^s$. Let

$$\mathcal{C}_1 = \{F \cdot T : \quad F \in \mathcal{F}(E) \ \text{ and } \ T \in \mathcal{T}(E)\}.$$

For every element $E_1 \in \mathcal{C}_1$, one can fix θ_1 such that $E_1 = E(\theta_1)$. Let $K_1 = K(\theta_1)$. Then K_1 is a finite extension of K and E_1/K_1 is abelian. By Corollary 2.3 and the above claim, there are a finite set $\mathcal{T}(E_1)$ consisting of all finite extensions of E_1 with the degree dividing t and a finite set $\mathcal{F}(E_1)$ of all covering F_1's of E_1 with respect to K_1 such that $Gal(F_1/E_1)$ are the elementary abelian p-groups with $\#Gal(F_1/E_1) \leq p^s$. Let

$$\mathcal{C}_2 = \bigcup_{E_1 \in \mathcal{C}_1} \mathcal{F}(E_1) \cdot \mathcal{T}(E_1)$$

where

$$\mathcal{F}(E_1)\cdot\mathcal{T}(E_1)=\{F_1\cdot T_1:\quad F_1\in\mathcal{F}(E_1)\ \text{ and }\ T_1\in\mathcal{T}(E_1)\}.$$

For every element $E_2\in\mathcal{F}(E_1)\cdot\mathcal{T}(E_1)$, one can fix θ_2 such that $E_2=E_1(\theta_2)$. Let $K_2=K_1(\theta_2)$. Then K_2 is a finite extension of K and E_2/K_2 is abelian. By Corollary 2.3 and the above claim, there are a finite set $\mathcal{T}(E_2)$ consisting of all finite extensions of E_2 with the degree dividing t and a finite set $\mathcal{F}(E_2)$ of all covering F_2's of E_2 with respect to K_2 such that $Gal(F_2/E_2)$ are the elementary abelian p-group with $\#Gal(F_2/E_2)\le p^s$. Let

$$\mathcal{C}_3=\bigcup_{E_1\in\mathcal{C}_1}\ \bigcup_{E_2\in\mathcal{F}(E_1)\cdot\mathcal{T}(E_1)}\mathcal{F}(E_2)\cdot\mathcal{T}(E_2)$$

where

$$\mathcal{F}(E_2)\cdot\mathcal{T}(E_2)=\{F_2\cdot T_2:\quad F_2\in\mathcal{F}(E_2)\ \text{ and }\ T_2\in\mathcal{T}(E_2)\}.$$

.....

Inductively, for every element $E_{k-1}\in\mathcal{F}(E_{k-2})\cdot\mathcal{T}(E_{k-2})$, one can fix θ_{k-1} such that $E_{k-1}=E_{k-2}(\theta_{k-1})$. Let $K_{k-1}=K_{k-2}(\theta_{k-1})$. Then K_{k-1} is a finite extension of K and E_{k-1}/K_{k-1} is abelian. By Corollary 2.3 and the above claim, there is a finite set $\mathcal{T}(E_{k-1})$ consisting of all finite extensions of E_{k-1} with the degree dividing t and a finite set $\mathcal{F}(E_{k-1})$ of all covering F_{k-1}'s of E_{k-1} with respect to K_{k-1} such that $Gal(F_{k-1}/E_{k-1})$ are the elementary abelian p-group with $\#Gal(F_{k-1}/E_{k-1})\le p^s$. Let

$$\mathcal{C}_k=\bigcup_{E_1\in\mathcal{C}_1}\ \bigcup_{E_2\in\mathcal{F}(E_1)\cdot\mathcal{T}(E_1)}\cdots\bigcup_{E_{k-1}\in\mathcal{F}(E_{k-2})\cdot\mathcal{T}(E_{k-2})}\mathcal{F}(E_{k-1})\cdot\mathcal{T}(E_{k-1})$$

where

$$\mathcal{F}(E_{k-1})\cdot\mathcal{T}(E_{k-1})=\{F_{k-1}\cdot T_{k-1}:\quad F_{k-1}\in\mathcal{F}(E_{k-1})\ \text{ and }\ T_{k-1}\in\mathcal{T}(E_{k-1})\}.$$

Suppose L is a k-covering of E with respect to K. Let $L=E(\eta)$ and

$$\sigma\eta=\sum_{i=0}^{k-1}a_i(\sigma)\eta^i$$

for all $\sigma\in Gal(L/E)$. Let K' be a finite extension of K obtained by adjoining all $a_i(\sigma)$'s for $0\le i\le k-1$ and $\sigma\in Gal(L/E)$ and all coefficients of the minimal polynomial of η over E to K. Then

$$Gal(K'(\eta)/K')=Gal(L/E)$$

which is solvable by Corollary 5 of [8] Chapter IV §2.

Let L' be the maximal abelian extension of E inside L. Then L' is a nontrivial finite covering of E with respect to K by the maximality. One can decompose $Gal(L'/E)$ into the p-part and the non-p-part and both parts are the finite coverings of E with respect to K. For the p-part, one can further obtain a non-trivial covering Σ of E with respect to K such that $Gal(\Sigma/E)$ is a p-elementary abelian group. By

composing the non-p-part with Σ, one obtains $L_1 \in \mathcal{C}_1$ such that L_1 is a non-trivial finite covering of E with respect to K and $L_1 \subseteq L'$.

If $L \neq L_1$, there is the non-trivial maximal abelian extension L_1' of L_1 inside L. Then L_1' is a finite covering of L_1 with respect to K. By the same argument as above, one can obtain $L_2 \in \mathcal{C}_2$ such that L_2 is a non-trivial finite covering of L_1 with respect to K and $L_2 \subseteq L_1'$. $\cdots$. By induction, one concludes that

$$L \in \bigcup_{i=1}^{k} \mathcal{C}_i.$$

Step 2. In general, let $E' = E(\mu_{p^\infty})$. Then $Gal(E'/E)$ is a closed subgroup of $\mathbb{Z}_p^\times$ and the set of the intermediate fields

$$\mathcal{E} = \{E \subseteq E_1 \subseteq E' : \ [E_1 : E] | k\}$$

is finite. For each $E_1 \in \mathcal{E}$, one can write $k_1 = k/([E_1 : E])$. By Step 1, there is a finite set $\mathcal{G}(E_1)$ consisting of all k_1-coverings Γ' of E' with respect to K such that the group extension

$$1 \to Gal(\Gamma'/E') \to Gal(\Gamma'/E_1) \to Gal(E'/E_1) \to 1$$

is split. Let $Gal(E'/E_1)^{k_1}$ be the closed subgroup of $Gal(E'/E_1)$ generated by the k_1-th power of all elements in $Gal(E'/E_1)$. Since $Gal(E'/E_1)$ is a closed subgroup of $\mathbb{Z}_p^\times$, the quotient group $Gal(E'/E_1)/Gal(E'/E_1)^{k_1}$ is finite. For each $\Gamma' \in \mathcal{G}(E_1)$, the closed subgroups of $Gal(\Gamma'/E_1)$ with index k_1 is bounded by the number subgroups of $Gal(\Gamma'/E') \times [Gal(E'/E_1)/Gal(E'/E_1)^{k_1}]$. This implies that the set of intermediate fields

$$\mathcal{H}(\Gamma') = \{E_1 \subseteq \Gamma \subseteq \Gamma' : \ [\Gamma : E_1] = k_1\}$$

is finite. Therefore

$$\mathcal{H} = \bigcup_{E_1 \in \mathcal{E}} \bigcup_{\Gamma' \in \mathcal{G}(E_1)} \mathcal{H}(\Gamma')$$

is finite. For any k-covering F of E with respect to K, one has $F \cap E' \in \mathcal{E}$ and $F \in \mathcal{H}$. The proof is complete. □

6. The explicit computation

In this section, we will compute the number of p-covering of two extreme cases $\mathbb{Q}_p(\mu_{p^\infty})$ and $\mathbb{Q}_p^{(1)}$ with respect to $\mathbb{Q}_p$. First we need some results from $\phi - \Gamma -$ modules described in A3,3.1 of [2]. Let A be the integral ring of the algebraic closure of $\mathbb{Q}_p$ and

$$R = \varprojlim_{n \in \mathbb{N}} (A/pA)$$

where the transition map is $a \mapsto a^p$.

For any $x = (\widetilde{x_n})_{n \geq 0} \in R$, one can show that $v_p(x_n^{p^n})$ is stable if n is sufficient large where v_p is the normalized p-adic valuation on A with $v_p(p) = 1$ and x_n is

any lifting of $\widetilde{x_n}$ to A. Moreover $v(x) = v_p(x_n^{p^n})$ for sufficient large n defines a valuation on R and R is a complete valuation ring with respect to this valuation. The fraction field $Fr\ R$ is algebraic closed and of characteristic p.

Fix the set $\{\zeta_{p^n}\}_{n\geq 0}$ of the p^n-th primitive roots of unity satisfying $\zeta_{p^{n+1}}^p = \zeta_{p^n}$ for $n \geq 1$. Set $\varpi_n = \zeta_{p^n} - 1$ and $\varpi = (\varpi_n \mod p)_{n\geq 0} \in R$. Then $v(\varpi) > 0$ and $\mathbb{F}_p[[\varpi]] \subseteq R$ where $\mathbb{F}_p$ is the finite field of p elements. By 3.1.6 Theoreme in [2], the natural action of $Gal(\bar{\mathbb{Q}}_p/\mathbb{Q}_p)$ on the separable closure $\mathbb{F}_p((\varpi))^{sep}$ of $\mathbb{F}_p((\varpi))$ in FrR is stable. Moreover, this action induces the isomorphism of the Galois groups

$$\Phi: \quad Gal[\bar{\mathbb{Q}}_p/\mathbb{Q}_p(\mu_{p^\infty})] \longrightarrow Gal[\mathbb{F}_p((\varpi))^{sep}/\mathbb{F}_p((\varpi))].$$

For a positive integer i with $(p,i) = 1$ and $a \in \mathbb{F}_p[[\varpi]]^\times$, one can consider the Artin-Schreier extension $F_{i,a}$ of $\mathbb{F}_p((\varpi))$ inside $\mathbb{F}_p((\varpi))^{sep}$ given by the equation $y^p - y - a\varpi^{-i} = 0$. Let $G_{i,a} = \Phi^{-1}(Gal[\mathbb{F}_p((\varpi))^{sep}/F_{i,a}])$. The following proposition gives the corresponding finite extension of $\mathbb{Q}_p(\mu_{p^\infty})$ inside $\bar{\mathbb{Q}}_p$ by $G_{i,a}$.

Proposition 6.1. *Let $a = (\widetilde{a_n})_{n\geq 0} \in \mathbb{F}_p[[\varpi]]^\times$ where a_n be a lifting of $\widetilde{a_n}$ in $\mathbb{Q}_p(\varpi_n)$ for all $n \geq 0$. If $p^{n-1} > i$, then*

$$\bar{\mathbb{Q}}_p^{G_{i,a}} = \mathbb{Q}_p(\mu_{p^\infty}, \sqrt[p]{1 + a_n\theta^p(\varpi_{n+1}^p)^{-i}})$$

where $\theta \in \mathbb{Q}_p(\varpi_1)$ and $\theta^{p-1} + p = 0$.

Proof. Let x_n be a root of the equation

$$X^p - (\varpi_{n+1}^p)^{i(p-1)}X - a_n(\varpi_{n+1}^p)^{i(p-1)} = 0.$$

Then

$$v_p(x_n) = \frac{i(p-1)}{p}v_p((\varpi_{n+1}^p)) = \frac{i}{p^n}$$

for all $n \geq 1$. If $p^{n-1} > i$, the other roots $x_n(\alpha)$ of the above equation satisfy

$$x_n(\alpha) \equiv x_n + \alpha(\varpi_{n+1}^p)^i \mod p$$

for $\alpha \in \mu_{p-1}$ by Hensel's Lemma (see Chapter II, (1.3) Corollary 3 of [4]). Let λ_{n+1} be a root of the equation by replacing n with $n+1$ in the above equation and $\lambda_{n+1}(\alpha)$ be the other roots satisfying

$$\lambda_{n+1}(\alpha) \equiv \lambda_{n+1} + \alpha(\varpi_{n+2}^p)^i \mod p$$

for $\alpha \in \mu_{p-1}$. Since $\varpi_{n+2}^{p^2} \equiv \varpi_{n+1}^p \mod p^2$, one has

$$\begin{aligned}
&(\lambda_{n+1}^p - x_n)\prod_{\alpha\in\mu_{p-1}}(\lambda_{n+1}^p - x_n(\alpha))\\
&= (\lambda_{n+1}^p)^p - (\varpi_{n+1}^p)^{i(p-1)}\lambda_{n+1}^p - a_n(\varpi_{n+1}^p)^{i(p-1)}\\
&= [(\varpi_{n+2}^p)^{i(p-1)}\lambda_{n+1} + a_{n+1}(\varpi_{n+2}^p)^{i(p-1)}]^p - (\varpi_{n+1}^p)^{i(p-1)}\lambda_{n+1}^p - a_n(\varpi_{n+1}^p)^{i(p-1)}\\
&\equiv (\varpi_{n+1}^p)^{i(p-1)}(\lambda_{n+1} + a_{n+1})^p - (\varpi_{n+1}^p)^{i(p-1)}\lambda_{n+1}^p - a_n(\varpi_{n+1}^p)^{i(p-1)} \mod p^2\\
&\equiv 0 \mod p(\varpi_{n+1}^p)^{i(p-1)}
\end{aligned}$$

for $p^{n-1} > i$.

If $v_p(\lambda_{n+1}^p - x_n) > v_p((\varpi_{n+1}^p)^i)$, then

$$v_p(\lambda_{n+1}^p - x_n(\alpha)) = v_p(\lambda_{n+1}^p - x_n + x_n - x_n(\alpha)) = v_p((\varpi_{n+1}^p)^i)$$

for all $\alpha \in \mu_{p-1}$. Therefore $\lambda_{n+1}^p \equiv x_n \mod p$.

Otherwise there is $\alpha_0 \in \mu_{p-1}$ such that $v_p(\lambda_{n+1}^p - x_n(\alpha_0)) > v_p((\varpi_{n+1}^p)^i)$. By the same argument as above, one has $\lambda_{n+1}^p \equiv x_n(\alpha_0) \mod p$. Let

$$\alpha_0' = \begin{cases} \alpha_0 & \text{if } p = 2 \\ -\alpha_0 & \text{if } p > 2. \end{cases}$$

Then $\alpha_0' \in \mu_{p-1}$ and

$$\lambda_{n+1}(\alpha_0')^p \equiv (\lambda_{n+1} + \alpha_0'(\varpi_{n+2}^p)^i)^p \equiv \lambda_{n+1}^p + (\alpha_0'(\varpi_{n+2}^p)^i)^p \equiv x_n \mod p.$$

Set

$$x_{n+1} = \begin{cases} \lambda_{n+1} & \text{if } v_p(\lambda_{n+1}^p - x_n) > v_p((\varpi_{n+1}^p)^i) \\ \lambda_{n+1}(\alpha_0') & \text{otherwise} \end{cases}$$

for $p^{n-1} > i$. If $p^{n-1} \leq i$, we set $x_{n-1} = x_n^p$. Then $x = (x_n \mod p)_{n\geq 0} \in R$ and $y_0 = \varpi^{-i}x$ is a root of the equation $y^p - y - a\varpi^{-i} = 0$ over $\mathbb{F}_p((\varpi))$.

If $p^{n-1} > i$, one has

$$\begin{aligned} v_p[(\varpi_{n+2}^p)^{i(p-1)}(x_{n+1}(\alpha) - x_{n+1})] &= ipv_p(\varpi_{n+2}^p) \\ <1 \leq v_p(x_{n+1}^p - x_n) &= v_p[(\varpi_{n+2}^p)^{i(p-1)}(x_{n+1} + a_{n+1}) - x_n] \end{aligned}$$

for all $\alpha \in \mu_{p-1}$. By Krasner's Lemma (see Chapter VIII, (8.1.6) of [7]), one concludes that

$$\mathbb{Q}_p(\mu_{p^\infty}, x_{n+1}) \subseteq \mathbb{Q}_p(\mu_{p^\infty}, x_n)$$

for $p^{n-1} > i$. Therefore $Gal[\bar{\mathbb{Q}}_p/\mathbb{Q}_p(\mu_{p^\infty}, x_n)]$ fixes y_0. By the isomorphism Φ, one concludes that $\bar{\mathbb{Q}}_p^{G_{i,a}} = \mathbb{Q}_p(\mu_{p^\infty}, x_n)$.

Since $p^{n-1} \geq 1 + i \geq 1 + \frac{2i}{p}$, one has

$$v_p(\theta x_n^2(\varpi_{n+1}^p)^{-2i}) = \frac{1}{p-1} + \frac{2i}{p^n} - \frac{2i}{p^{n-1}(p-1)} = \frac{p^n - 2i}{p^n(p-1)} \geq \frac{1}{p^{n-1}(p-1)}.$$

This implies that

$$\begin{aligned} (1 + \theta x_n(\varpi_{n+1}^p)^{-i})^p &\equiv 1 + p\theta x_n(\varpi_{n+1}^p)^{-i} + (\theta x_n(\varpi_{n+1}^p)^{-i})^p \mod p\theta\varpi_n \\ &\equiv 1 + a_n\theta^p(\varpi_{n+1}^p)^{-i} \mod p\theta\varpi_n \end{aligned}$$

By Hensel's Lemma, one obtains that

$$\sqrt[p]{1 + a_n\theta^p(\varpi_{n+1}^p)^{-i}} \in \mathbb{Q}_p(\mu_{p^{n+1}}, x_n).$$

Finally one only needs to show that

$$\sqrt[p]{1 + a_n\theta^p(\varpi_{n+1}^p)^{-i}} \notin \mathbb{Q}_p(\mu_{p^\infty}).$$

Suppose not, one has

$$\sqrt[p]{1 + a_n\theta^p(\varpi_{n+1}^p)^{-i}} \in \mathbb{Q}_p(\mu_{p^{n+2}})$$

since $1 + a_n\theta^p(\varpi_{n+1}^p)^{-i} \in \mathbb{Q}_p(\mu_{p^{n+1}})$. There is $\xi \in \varpi_{n+2}\mathbb{Z}_p[\varpi_{n+2}]$ such that

$$(1+\xi)^p = 1 + a_n\theta^p(\varpi_{n+1}^p)^{-i}.$$

This implies that $\xi^p \equiv 0 \mod p$ and

$$a_n\theta^p(\varpi_{n+1}^p)^{-i} \equiv p\xi + \xi^p \mod 2p\xi^2.$$

Therefore there is $\kappa \in \varpi_{n+2}\mathbb{Z}_p[\varpi_{n+2}]$ such that

$$(\theta^{-1}\xi)^p - \theta^{-1}\xi - a_n(\varpi_{n+1}^p)^{-i} = \kappa.$$

Consider the equation

$$(z+\theta^{-1}\xi)^p - (z+\theta^{-1}\xi) - a_n(\varpi_{n+1}^p)^{-i} \equiv z^p - z \mod \varpi_{n+2}.$$

By Hensel's Lemma, one concludes that $x_n \in \mathbb{Q}_p(\mu_{p^{n+2}})$. A contradiction is derived. □

Let χ be the cyclotomic character which gives the isomorphism

$$\chi : \ Gal[\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p] \longrightarrow \mathbb{Z}_p^\times.$$

For any $\sigma \in Gal[\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p)]$, one has $\sigma(\varpi) = (\varpi+1)^{\chi(\sigma)} - 1$.

Let $\mathfrak{P} : \mathbb{F}_p((\varpi)) \longrightarrow \mathbb{F}_p((\varpi))$ be a homomorphism of the additive group by sending $x \mapsto x^p - x$. It is well-known that two Artin-Schreier extensions $y^p - y - \alpha$ and $y^p - y - \beta$ are the same in the fixed algebraic closure if and only if α and β generate the same group in $\mathbb{F}_p((\varpi))/\mathfrak{P}[\mathbb{F}_p((\varpi))]$. By Hensel's Lemma, all Artin-Schreier extensions of degree p are given by

$$y^p - y - \sum_{p\nmid k,\ 1\le k\le i} \alpha_k\varpi^{-k} - \alpha_0$$

where $\alpha_i \neq 0$ and $\alpha_k \in \mathbb{F}_p$ for $0 \le k \le i$.

By the isomorphism Φ, those Artin-Schreier extensions of degree p as above will produce the p-coverings of $\mathbb{Q}_p(\mu_{p^\infty})$ with respect to $\mathbb{Q}_p$ if and only if the subgroup generated by

$$\sum_{p\nmid k,\ 1\le k\le i} \alpha_k\varpi^{-k} + \alpha_0$$

in $\mathbb{F}_p((\varpi))/\mathfrak{P}[\mathbb{F}_p((\varpi))]$ is stable under the natural action of $Gal[\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p]$.

Lemma 6.2. *Suppose the subgroup generated by*

$$\sum_{p\nmid k,\ 1\le k\le i} \alpha_k\varpi^{-k} + \alpha_0$$

in $\mathbb{F}_p((\varpi))/\mathfrak{P}[\mathbb{F}_p((\varpi))]$ is stable under the natural action of $Gal[\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p]$.

If $p > 2$, then $i \le p-1$ or $i = 2p-1$ with $j\alpha_{2p-j} + (j+1)\alpha_{2p-j-1} = 0$ for $1 \le j \le p-2$.

If $p = 2$, then $i \le 5$.

Proof. First we consider $p > 2$. There is $\sigma_0 \in Gal[\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p]$ such that $\chi(\sigma_0) = 1 + p$. Then

$$\sigma_0(\varpi) = (1+\varpi)^{1+p} - 1 = (1+\varpi)(1+\varpi^p) - 1 = \varpi(1+\varpi^{p-1}(1+\varpi))$$

and

$$\sigma_0(\varpi^{-k}) = \varpi^{-k} - k\varpi^{p-1-k} + (\text{power of } \varpi \geq p-k)$$

for $1 \leq k \leq i$. Therefore

$$\begin{aligned}&\sigma_0(\varpi^{-i} + \sum_{p\nmid k,\ 1\leq k\leq i-1} \alpha_k \varpi^{-k} + \alpha_0) - (\varpi^{-i} + \sum_{p\nmid k,\ 1\leq k\leq i-1} \alpha_k \varpi^{-k} + \alpha_0)\\ &= -i\varpi^{p-1-i} + (\text{power of } \varpi \geq p-i).\end{aligned}$$

Suppose $i > p$. By the assumption, one concludes that $p - 1 - i \equiv 0 \mod p$. So one can write $i = np - 1$ with $n \geq 2$. Then

$$\begin{aligned}\sigma_0(\varpi^{-i}) &= \varpi^{-i}[1+\varpi^{p-1}(1+\varpi)][1+\varpi^{p(p-1)}(1+\varpi^p)]^{-n}\\ &= \varpi^{-i} + \varpi^{(1-n)p} + \varpi^{(1-n)p+1} + (\text{power of } \varpi \geq (p-1-n)p+1).\end{aligned}$$

Moreover

$$\begin{aligned}\sigma_0(\varpi^{-i+j}) &= \varpi^{-i+j}[1+\varpi^{p-1}(1+\varpi)]^{j+1}[1+\varpi^{p(p-1)}(1+\varpi^p)]^{-n}\\ &= \varpi^{-i+j} + (j+1)\varpi^{(1-n)p+j} + (j+1)\varpi^{(1-n)p+j+1}\\ &\quad + (\text{power of } \varpi \geq (2-n)p+j-1).\end{aligned}$$

for $1 \leq j \leq p-2$ and

$$\sigma_0(\varpi^{-i+j}) = \varpi^{-i+j} + (\text{power of } \varpi \geq (2-n)p)$$

for $j \geq p$. By the assumption, the coefficients of $\varpi^{(1-n)p+j}$ for $1 \leq j \leq p-2$ in

$$\sigma_0(\sum_{p\nmid k,\ 1\leq k\leq i} \alpha_k \varpi^{-k} + \alpha_0) - (\sum_{p\nmid k,\ 1\leq k\leq i} \alpha_k \varpi^{-k} + \alpha_0)$$

should be zero. This implies that $j\alpha_{i-j+1} + (j+1)\alpha_{i-j} = 0$ for $1 \leq j \leq p-2$.

If $n \geq 3$, one further has $1 - n > (1-n)p + p - 1$. Then $(p-1)\alpha_{i-p+2} = 0$ by considering the coefficient of $\varpi^{(1-n)p+p-1}$. A contradiction is derived by using the above equations.

Now we consider the case $p = 2$. One can choose $\sigma_0 \in Gal[\mathbb{Q}_2(\mu_{2^\infty})/\mathbb{Q}_2]$ such that $\chi(\sigma_0) = 1 + 4$. Then

$$\sigma_0(\varpi) = (1+\varpi)^{1+4} - 1 = (1+\varpi)(1+\varpi^4) - 1 = \varpi(1+\varpi^3(1+\varpi))$$

and

$$\begin{aligned}\sigma_0(\varpi^{-k}) &= \varpi^{-k}[1+\varpi^3(1+\varpi)][1+\varpi^6(1+\varpi^2)]^{-\frac{k+1}{2}}\\ &= \varpi^{-k} + (\varpi^{\frac{3-k}{2}})^2 + \varpi^{4-k} + (\text{power of } \varpi \geq 6-k)\end{aligned}$$

for the odd integer k with $1 \leq k \leq i$. Suppose $i > 5$. Then

$$4 - i < \frac{3-i}{2} \leq \frac{3-k}{2}.$$

By the assumption, the coefficient of ϖ^{4-i} in

$$\sigma_0(\sum_{2\nmid k,\ 1\leq k\leq i}\alpha_k\varpi^{-k}+\alpha_0)-(\sum_{2\nmid k,\ 1\leq k\leq i}\alpha_k\varpi^{-k}+\alpha_0)$$

should be zero. A contradiction is derived. □

One of the main result of this section is the following theorem.

Theorem 6.3. *The number of p-coverings of $\mathbb{Q}_p(\mu_{p^\infty})$ with respect to $\mathbb{Q}_p$ is $2p-1$.*

Proof. First we consider the case that $p>2$. Fix a primitive $(p-1)$-th roots of unity $\zeta_{p-1}\in\mathbb{Z}_p^\times$ and $\tau_0\in Gal[\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p]$ such that $\chi(\tau_0)=\zeta_{p-1}$. Then $Gal[\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p]$ is topologically generated by τ_0 and σ_0 in Lemma 6.2. Therefore one only needs to determine those elements in Lemma 6.2 such that the cyclic subgroups generated by these elements in $\mathbb{F}_p((\varpi))/\mathfrak{P}[\mathbb{F}_p((\varpi))]$ are stable under the natural action of τ_0 and σ_0.

For the stability by action of σ_0, one needs to further consider the cases $i=2p-1$ and $i=p-1$ by the argument in Lemma 6.2.

For $i=2p-1$ in Lemma 6.2, one only needs to consider the coefficient of ϖ^{-1} in

$$\sigma_0(\sum_{k\neq p,\ 0\leq k\leq 2p-1}\alpha_k\varpi^{-k})-(\sum_{k\neq p,\ 0\leq k\leq 2p-1}\alpha_k\varpi^{-k})$$

which is $\alpha_{2p-1}+(p-1)\alpha_{p+1}=0$ by the condition in Lemma 6.2 and the constant term $\alpha_{2p-2}+\alpha_{p-1}$ which should be zero.

For $i=p-1$ in Lemma 6.2, the constant term in

$$\sigma_0(\sum_{k=0}^{p-1}\alpha_k\varpi^{-k})-(\sum_{k=0}^{p-1}\alpha_k\varpi^{-k})$$

is α_{p-1} and is never equal to 0.

Therefore one remains to consider the stability by action of τ_0 for the elements in Lemma 6.2 with $i\leq p-2$ or $i=2p-1$ with $j\alpha_{2p-j}+(j+1)\alpha_{2p-j-1}=0$ for $1\leq j\leq p-2$ and $\alpha_{2p-2}+\alpha_{p-1}=0$.

Let g be $\zeta_{p-1} \mod p$ in $\mathbb{F}_p$ and

$$\tau_0(\varpi^{-j})=[(1+\varpi)^{\zeta_{p-1}}-1]^{-j}=g^{-j}\varpi^{-j}+\sum_{k=0}^{j-1}b_{j,k}\varpi^{-k}+(\text{power of }\varpi\geq 1)$$

for $j\geq 1$. By computing the derivative of $\tau_0(\varpi^{-j})$ with respect to ϖ, one has the following equation

$$(1+\varpi)\tau_0(\varpi^{-j})'=-jg[\tau_0(\varpi^{-j})+\tau_0(\varpi^{-j-1})].$$

By comparing the coefficients of ϖ^{-l} in the above equation for $1\leq l\leq j$, one has

$$(l-1)b_{j,l-1}+lb_{j,l}=jg(b_{j,l}+b_{j+1,l}) \tag{6.4}$$

where $b_{j,j} = g^{-j}$. Let

$$c_l = l \sum_{j=l}^{2p-1} (-1)^{j-l} b_{j,l} j^{-1}$$

for $1 + p \le l \le 2p - 1$. Then

$$c_l - c_{l-1} + g^{-(l-1)} = g(b_{l,l} + (-1)^{2p-1-l} b_{2p,l})$$

for $2 + p \le l \le 2p - 1$ by (6.4). Since

$$\tau_0(\varpi^{-1}) = g^{-1}\varpi^{-1}[1 - \frac{1}{2}(g-1)\varpi + (\text{power of } \varpi \ge 2)],$$

one has

$$\tau_0(\varpi^{-1-p}) = g^{-1-p}\varpi^{-1-p}[1 - \frac{1}{2}(g-1)\varpi + (\text{power of } \varpi \ge 2)]$$

and

$$\tau_0(\varpi^{-2p}) = g^{-2p}\varpi^{-2p}[1 - (g-1)^p\varpi^p + (\text{power of } \varpi \ge 2p)].$$

Then

$$b_{2p,p} = -(g-1)g^{-2} \quad \text{and} \quad b_{p+1,p} = -\frac{1}{2}(g-1)g^{-2}$$

and $b_{2p,l} = 0$ for $1 + p \le l \le 2p - 1$. Therefore $c_l = g^{-1}$ for $1 + p \le l \le 2p - 1$.

When $i = 2p - 1$, one needs to determine all α_k with $0 \le k \le 2p - 1$ and $k \ne p$ such that the coefficient of ϖ^{-l} for all $l \ge 0$ in

$$f_{2p-1}(\varpi) = g^{2p-1}\tau_0(\sum_{k\ne p,\ 0\le k\le 2p-1} \alpha_k \varpi^{-k}) - (\sum_{k\ne p,\ 0\le k\le 2p-1} \alpha_k \varpi^{-k})$$

is zero.

For $l > p$, the coefficient of ϖ^{-l} in $f_{2p-1}(\varpi)$ is

$$g(\sum_{k=l}^{2p-1} \alpha_k b_{k,l}) - \alpha_l = g[\sum_{k=l}^{2p-1} (-1)^{k-l} l k^{-1} \alpha_l b_{k,l}] - \alpha_l = g\alpha_l c_l - \alpha_l = 0$$

by the recurrence of α_k for $p + 1 \le k \le 2p - 1$.

Since

$$\sum_{j=p+1}^{2p-1} (-1)^{j-(p+1)} b_{j,p-1} j^{-1} = g(b_{2p,p} - b_{p+1,p}) = -\frac{1}{2}(g-1)g^{-1}$$

by (6.4), the coefficient of ϖ^{1-p} in $f_{2p-1}(\varpi)$ is

$$\begin{aligned} & g(\sum_{k=p+1}^{2p-1} \alpha_k b_{k,p-1} + \alpha_{p-1} b_{p-1,p-1}) - \alpha_{p-1} \\ =& g\alpha_{p+1}[\sum_{k=p+1}^{2p-1} (-1)^{k-(p+1)} k^{-1} b_{k,p-1}] + (g-1)\alpha_{p-1} \\ =& -\frac{1}{2}(g-1)\alpha_{p+1} + (g-1)\alpha_{p-1} = (g-1)(\alpha_{2p-2} + \alpha_{p-1}) = 0. \end{aligned}$$

For $1 < l < p-1$, the coefficient of ϖ^{-l} in $f_{2p-1}(\varpi)$ is

$$g\sum_{j=p+1}^{2p-1}\alpha_j b_{j,l} + g\sum_{j=l}^{p-1}\alpha_j b_{j,l} - \alpha_l.$$

This coefficient is zero if and only if

$$\alpha_l = \frac{g}{1-g^{1-l}}(\sum_{j=p+1}^{2p-1}\alpha_j b_{j,l} + \sum_{j=l+1}^{p-1}\alpha_j b_{j,l})$$

which is unique determined inductively.

Considering the coefficient of ϖ^{-1} in $f_{2p-1}(\varpi)$, one can write

$$\begin{aligned} &g\tau_0(\sum_{k\neq p,\ 0\leq k\leq 2p-1}\alpha_k\varpi^{-k}) - (\sum_{k\neq p,\ 0\leq k\leq 2p-1}\alpha_k\varpi^{-k}) \\ &= \eta\varpi^{-1} + (\text{power of } \varpi \geq 0) \end{aligned}$$

for some $\eta \in \mathbb{F}_p$. Let τ_0 act on the above equation and one obtains

$$\begin{aligned} &g^2\tau_0^2(\sum_{k\neq p,\ 0\leq k\leq 2p-1}\alpha_k\varpi^{-k}) - g\tau_0(\sum_{k\neq p,\ 0\leq k\leq 2p-1}\alpha_k\varpi^{-k}) \\ &= \eta\varpi^{-1} + (\text{power of } \varpi \geq 0). \end{aligned}$$

$$\cdots\ \cdots\ \cdots$$

Inductively, one has

$$\begin{aligned} &g^{p-1}\tau_0^{p-1}(\sum_{k\neq p,\ 0\leq k\leq 2p-1}\alpha_k\varpi^{-k}) - g^{p-2}\tau_0^{p-2}(\sum_{k\neq p,\ 0\leq k\leq 2p-1}\alpha_k\varpi^{-k}) \\ &= \eta\varpi^{-1} + (\text{power of } \varpi \geq 0). \end{aligned}$$

Since $g^{p-1} = 1$ and $\tau_0^{p-1} = 1$, one concludes that $(p-1)\eta = 0$. This implies that η is always zero and α_1 can be any element in $\mathbb{F}_p$.

The constant term of $f_{2p-1}(\varpi)$ is zero if and only if

$$\alpha_0 = \frac{g}{1-g}(\sum_{k=p+1}^{2p-1}\alpha_k b_{k,0} + \sum_{k=1}^{p-1}\alpha_k b_{k,0})$$

which is uniquely determined once α_1 is fixed.

Therefore one concludes that there are p subgroups generated by such $f_{2p-1}(\varpi)$ in $\mathbb{F}_p((\varpi))/\mathfrak{P}[\mathbb{F}_p((\varpi))]$ which are stable under the natural action of $Gal[\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p]$.

When $i \leq p-2$, one needs to determine all α_k with $0 \leq k \leq p-2$ such that the coefficient of ϖ^{-l} for all $l \geq 0$ in

$$f_i(\varpi) = g^i\tau_0(\sum_{k=0}^{i}\alpha_k\varpi^{-k}) - (\sum_{k=0}^{i}\alpha_k\varpi^{-k})$$

is zero. This is equivalent to

$$\alpha_l = \frac{g^i}{1-g^{i-l}}\sum_{k=l+1}^{i}\alpha_k b_{k,l}$$

which is unique determined inductively. Therefore one concludes that there is one subgroup generated by $f_i(\varpi)$ in $\mathbb{F}_p((\varpi))/\mathfrak{P}[\mathbb{F}_p((\varpi))]$ which is stable under the natural action of $Gal[\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p]$.

Now we consider the case of $p = 2$. There is $\tau_0 \in Gal[\mathbb{Q}_2(\mu_{2^\infty})/\mathbb{Q}_2]$ such that $\chi(\tau_0) = -1$. Then $Gal[\mathbb{Q}_2(\mu_{2^\infty})/\mathbb{Q}_2]$ is topologically generated by τ_0 and σ_0 in Lemma 6.2. Since

$$\sigma_0(\varpi^{-5}) - \varpi^{-5}, \quad \sigma_0(\varpi^{-3}) - \varpi^{-3} - 1 \quad \text{and} \quad \sigma_0(\varpi^{-1}) - \varpi^{-1}$$

and

$$\tau_0(\varpi^{-k}) - \varpi^{-k} - 1 \quad \text{for } k = 1, 3, 5$$

are all in $\mathfrak{P}[\mathbb{F}_p((\varpi))]$ by the direct computation, one concludes that only the following three subgroups generated by

$$1, \quad \varpi^{-1} + \varpi^{-5}, \quad 1 + \varpi^{-1} + \varpi^{-5}$$

in $\mathbb{F}_p((\varpi))/\mathfrak{P}[\mathbb{F}_p((\varpi))]$ respectively are stable under the action of $Gal[\mathbb{Q}_2(\mu_{2^\infty})/\mathbb{Q}_2]$. The proof is complete. □

By modifying the above argument, one can further compute the number of p-covering of $\mathbb{Q}_p^{(1)}$ with respect to $\mathbb{Q}_p$. First of all, one needs to modify Φ to the isomorphism of the Galois groups

$$\bar{\Phi}: \quad Gal[\bar{\mathbb{Q}}_p/\mathbb{Q}_p^{(1)}] \longrightarrow Gal[\bar{\mathbb{F}}_p((\varpi))^{sep}/\bar{\mathbb{F}}_p((\varpi))]$$

inducing by the natural action of $Gal[\bar{\mathbb{Q}}_p/\mathbb{Q}_p^{(1)}]$ on FrR by 3.1.6 Theoreme in [2]. Then all Artin-Schreier extensions of degree p are given by

$$y^p - y - \sum_{p\nmid k,\ 1\le k\le i} \alpha_k \varpi^{-k}$$

where $\alpha_i \neq 0$ and $\alpha_k \in \bar{\mathbb{F}}_p$ for $0 \le k \le i$.

By the isomorphism $\bar{\Phi}$, those Artin-Schreier extensions of degree p as above will produce the p-coverings of $\mathbb{Q}_p^{(1)}$ with respect to $\mathbb{Q}_p$ if and only if the subgroup generated by

$$\sum_{p\nmid k,\ 1\le k\le i} \alpha_k \varpi^{-k}$$

in $\bar{\mathbb{F}}_p((\varpi))/\mathfrak{P}[\bar{\mathbb{F}}_p((\varpi))]$ is stable under the natural action of $Gal(\mathbb{Q}_p^{(1)}/\mathbb{Q}_p)$. Moreover the corresponding Lemma 6.2 still holds.

Theorem 6.5. *There are $(p-1)(p^2+p-1)$ p-coverings of $\mathbb{Q}_p^{(1)}$ with respect to $\mathbb{Q}_p$ for $p > 2$ and 7 for $p = 2$.*

Proof. We keep the same notation as that in Thm.6.3. The significant difference between this computation with that in Thm.6.3 is that one does not need to consider the constant terms but has to consider the Frobenius action.

For $p > 2$, the cyclic subgroups in $\bar{\mathbb{F}}_p((\varpi))/\mathfrak{P}[\bar{\mathbb{F}}_p((\varpi))]$ generated by those elements in Lemma 6.2 are stable under the natural action of σ_0 by the argument in Thm.6.3. Moreover, the cyclic subgroups in $\bar{\mathbb{F}}_p((\varpi))/\mathfrak{P}[\bar{\mathbb{F}}_p((\varpi))]$ generated by

$$\sum_{k\neq p,\ 1\leq k\leq 2p-1} \alpha_k \varpi^{-k} \qquad \text{with} \qquad \alpha_{2p-1} \neq 0$$

are stable under the natural action of τ_0 if and only if α_k with $1 < k < p-1$ are uniquely determined by

$$\alpha_k = \frac{g}{1-g^{1-k}}\Big(\sum_{j=p+1}^{2p-1} \alpha_j b_{j,k} + \sum_{j=k+1}^{p-1} \alpha_j b_{j,k}\Big).$$

Since these subgroups should be stable under the Frobenius action as well, one has $\alpha_k^p = c\alpha_k$ with $c \in \mathbb{F}_p^\times$ for all $1 \leq k \leq 2p-1$ and $k \neq p$. For any fixed $c \in \mathbb{F}_p^\times$, there are $p-1$ elements of α_{2p-1} and p elements of α_k with $2p-1 < k < p$ or $k = 1$ or $p-1$ satisfying the equation $x^p = cx$. Therefore one concludes that there are $(p-1)p^2$ subgroups in $\bar{\mathbb{F}}_p((\varpi))/\mathfrak{P}[\bar{\mathbb{F}}_p((\varpi))]$ generated by the elements of the above type which are stable under the natural action of $Gal(\mathbb{Q}_p^{(1)}/\mathbb{Q}_p)$.

For $1 \leq i \leq p-1$ with $\alpha_i \neq 0$, the coefficient of ϖ^{-l} in

$$g^i \tau_0\Big(\sum_{k=1}^{i} \alpha_k \varpi^{-k}\Big) - \Big(\sum_{k=1}^{i} \alpha_k \varpi^{-k}\Big)$$

is zero if and only if

$$\alpha_l = \frac{g^i}{1-g^{i-l}} \sum_{k=l+1}^{i} \alpha_k b_{k,l}$$

for $1 \leq l \leq i-1$. By considering the Frobenius action, there is $c \in \mathbb{F}_p^\times$ such that $a_i^p = ca_i$. One concludes that there are $p-1$ subgroups generated by

$$\sum_{k=1}^{i} \alpha_k \varpi^{-k} \quad \text{in} \quad \bar{\mathbb{F}}_p((\varpi))/\mathfrak{P}[\bar{\mathbb{F}}_p((\varpi))]$$

which are stable under the natural action of $Gal(\mathbb{Q}_p^{(1)}/\mathbb{Q}_p)$ for each $1 \leq i \leq p-1$.

For $p = 2$, one has $k \leq 5$ for the subgroups generated by

$$\sum_{2\nmid k,\ 1\leq k\leq i} \alpha_k \varpi^{-k}$$

in $\bar{\mathbb{F}}_2((\varpi))/\mathfrak{P}[\bar{\mathbb{F}}_2((\varpi))]$ which are stable under the natural action of σ_0 and τ_0 by Lemma 6.2. By considering the Frobenius action, one obtains that $\alpha_k \in \mathbb{F}_2$. Since there is at least one of α_k with $k = 1, 3, 5$ in $\mathbb{F}_2^\times$, one concludes that there are 7 such subgroups which are stable under the natural action of $Gal(\mathbb{Q}_p^{(1)}/\mathbb{Q}_p)$. The proof is complete. □

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